

Neural networks are *parametric models* because they have a fixed set of adjustable parameters. A simple polynomial function like $f(x) = ax^3 + bx^2 + c$ is a parametric model because we have three parameters (a,b,c) that we can train to fit this function to some data.

A *nonparametric model* is a model that either has no trainable parameters or has the ability to dynamically adjust the number of parameters it has, based on the training data. Kernel regression is an example of a nonparametric model for prediction; the simplest version of kernel regression is to simply find the nearest x_i points in the training data, X, to some new input, x, and then return the corresponding $y \in Y$ in the training data that is the average (figure 10.9).

Figure 10.9. One way to perform nonparametric kernel regression to predict the y component of a new x value is to find the most similar (i.e., the closest) x's in the training data and then take the average of their respective y components.